Week 1: Introduction to HPC

Overview

Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8:Task parallelism and manager worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Week 1: Introduction to HPC

- This week provides an introduction to high performance computing
- We will look at the background to the area and some example application areas
- Understanding hardware is important for parallel programming in achieving high performance so we will look at
    - trends in processor design
    - the cluster architecture

Syllabus plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

Week 1 delivery

- Overview
- Lecture units 1.1-1.5
- Summary
- Week 1 Quiz
- Week 1: Lecture 1 (16th January) and Lecture 2 (16th January)

Lecture Units

- Unit 1.1: What is HPC and what is it used for?
- Unit 1.2: The Top500 list
- Unit 1.3: The need for parallelism
- Unit 1.4: The cluster architecture
- Unit 1.5: HPC languages

Workshop 1

- Workshop 1 will be on Tuesday 20th of January and Wednesday 21st of January (attend your personalised session)
- In Workshop 1 you will be using the Top 500 list to explore trends in supercomputing
- Before the workshop you will need to have viewed units 1.1-1.3